Sahaj Bhadja

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Summary

Software engineer with 2+ years building **AI-powered applications**, **backend APIs on AWS**, and **data processing pipelines**. Experienced with prompt engineering, LLM integration (GPT-4, Claude), and modern cloud-native development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **AI applications integrating GPT-4o, Claude, and vision APIs** with prompt engineering techniques, optimizing prompts for improved accuracy and implementing switchable model architectures for 10K+ daily requests.
- Developed **REST APIs and backend services** with Python/Flask, implementing data workflows that processed content and reduced feedback turnaround by **60%** through AI-assisted automation.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **backend services on cloud infrastructure** (Azure) serving 500+ users, implementing API endpoints and data processing workflows that reduced latency from **1.2s to 350ms** through optimization.
- Developed **data pipelines** processing 5K+ monthly jobs with SQL databases, validation, and error handling, achieving 99%+ reliability while collaborating with data engineers and product teams.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **AI/ML experimentation pipelines** processing 100K+ records using PyTorch, implementing feature extraction and model training workflows with experiment tracking for reproducible results.
- Applied **analytical thinking to AI model optimization**, understanding model strengths, limitations, and context windows to improve planning accuracy by 18% through systematic experimentation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **data workflows using AWS Glue and S3**, implementing ETL pipelines with event-driven patterns and improving pipeline efficiency by **68%** through automation and optimization.
- Built **REST APIs** and maintained automated test suites using Python, participating in code reviews and adhering to modern engineering standards in Agile/Scrum sprints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **backend services with SQL/NoSQL databases** (Azure SQL, MongoDB), implementing APIs for 8 microservices handling 2M+ records with data validation and processing workflows.
- Built **frontend components using modern JavaScript frameworks** (React), implementing responsive interfaces with API integration and real-time data synchronization via WebSockets.

Skills

AI/Prompt Engineering: GPT-4/Claude APIs, prompt optimization, AI-assisted development, LLM integration
Backend & Cloud: Python, REST APIs, AWS (S3, Glue, Lambda basics), Azure, serverless patterns
Databases: PostgreSQL, MySQL, MongoDB, SQL queries, data pipelines, event-driven workflows
Frontend: React, Vue basics, TypeScript, JavaScript, modern JS frameworks
Practices: Git, CI/CD, Agile/Scrum, code reviews, clean code, security awareness

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built an **AI-powered content platform** with prompt engineering for document Q&A, achieving 95% query accuracy through optimized prompts and retrieval-augmented generation across 50K+ documents.

Linguistix Code | *Python, AWS (ECS, Fargate, CodePipeline), Docker*

- Built a **cloud-native API on AWS** using serverless patterns with ECS/Fargate, implementing automated deployment pipelines and validation gates for production-grade reliability.

PiXMod Live | *Next.js, MongoDB, Transformers, CLIP*

- Built an **AI application using transformer models** (CLIP, Stable Diffusion) for image processing, implementing semantic search and AI-driven automation workflows across 100K+ embeddings.